FalseGreen

VERIFICATION · SIGNED EVIDENCE

FalseGreen Verification Report

ACCEPTED

All required criteria passed for the exact source tree built and verified by FalseGreen.

This does not prove universal correctness. Assurance is limited to the explicit formal properties, assumptions, models, summaries, and scope recorded in this report.

THE WORK

Job	`301e97ff-620a-43c5-981c-57cf719b8c5d`
Language	rust
Toolchain	rustc 1.99.1 (rustc) (1.99.1)
Verifier environment	rust - rustc 1.99.1 (rustc) (1.99.1)

THE FROZEN BOUNDARY

Source tree	`1740a353282fe87b0dd82d8ead9f84dc963e98a719ab99313824b44155f8b610`
Contract SHA-256	`93cb341daba1ef96c3fee2f102e03244c355b2d1dfe6f8207b186ffc3cbf4919`
Criteria SHA-256	`04b8ab0a890de6a56a6403723bf51a2a46c5be704f5cfb3a85eaac89bc0528b3`
Criteria	4 / 4 PASSED

WHAT FALSEGREEN DID

Attempts	1
Duration	1m 11s
Report schema	v13
Report record	1 (the verdict mint)
Signed	2026-10-27 21:21:06 UTC

Contract (frozen)

Wall budget 2700s

Criterion	Title	Required	Result	Duration
`build`	the program compiles (release)	yes	**PASS**	20.3s
`tests`	the full test suite passes offline	yes	**PASS**	8.6s
`clippy`	clippy-clean (-D warnings)	yes	**PASS**	4.0s
`formatting`	rustfmt-clean	no	**PASS**	0.1s

Invariants chain linking is exact successor arithmetic and never wraps at the u64 ceiling — `formal-proofs, floor-green, both-rungs-proven`

Plan (frozen)

Objective	Verify the open-source falsegreen-ledger program exactly as published: the mechanical floor (release build, full offline test suite, clippy -D warnings, rustf mt) as frozen criteria, and one frozen claim — chain linking is exact successor arithmetic and never wraps at the u64 ceiling — carried by two formal encodings the machines discharge (Kani bounded over the shipped src/rules.r s arithmetic; an unbounded Creusot Why3 proof of the guarded Pearlite spec in src/spec.rs over both translated items).
Plan digest	`072290bc3d6605b2eb4a02abfc6f152c485f412abc5268815dd23477e1bfa8fd`
Work items	2

Work items, in dependency order:

Work item	Outcome	Depends on	Omission risk
`mechanical-floor`	the program builds, tests, lints and formats clean offline, exactly as published	-	a proof-only claim carried without the mechanical floor would leave the shipped binary itself unverified
`formal-proofs`	the chain-link claim is discharged by both formal encodings on the published source	mechanical-floor	the encodings encode the claim as read at freeze time — the encodingclaim equivalence is the frozen human judgement the report exposes verbatim

Acceptance conditions, and the evidence that decides each:

Scope	Condition	Evidence
`mechanical-floor`	every frozen criterion passes on the published source: release build, the full test suite, clippy with -D warnings, rustfmt check	build, behavioral-test, static-analysis, build
`formal-proofs`	the Kani rung returns proven over the shipped arithmetic (no counterexample), and the Creusot rung closes every verification condition for the frozen targets (measured: 4/4 obligations, incl. the shipped rule's u64-addition safety obligation)	proof, proof
`overall`	the published program is accepted: all four criteria green AND both proof rows proven, in one attempt, with the per-obligation prover disclosure and verbatim clauses in the signed report	behavioral-test, proof

Evidence manifest

ATTEMPT 1 - PASSED - 2026-10-27 21:21:06 UTC - WALL 1M 11S

source sha256: 1740a353282fe87b0dd82d8ead9f84dc963e98a719ab99313824b44155f8b610

env sha256: a4a5bf077e04f6aeb04a2ea1c0b0852414491d85daa6909080ad17cbf5f7f2e8

Security Findings

The bar these findings are judged against (the frozen policy): **fail on:** critical, high · **unknown severity:** insufficient_evidence.

Nothing to act on — 177 recorded facts (are inventory evidence in the signed JSON report).

The full SBOM and VEX are in the signed JSON report.

JSON report: https://falsegreen.com/report/301e97ff-620a-43c5-981c-57cf719b8c5d/json

Invariant Checks

Each entry is ONE frozen claim, the PROVER that worked on it, and the formal proposition that prover was asked to discharge — read out of the frozen encoding, so the machine's text sits next to the author's sentence. The full rows (any counterexample trace, the admitted assumptions) are in the signed JSON report.

ATTEMPT #1

chain linking is exact successor arithmetic and never wraps at the u64 ceiling

PROVEN

"record N links to record N-1 iff prev_hash equals record N-1's hash AND seq is the exact u64 successor of record N-1's seq; the successor is never taken modulo the u64 ceiling — no seq may link past u64::MAX (the chain cannot wrap) — and on the guarded domain (prev below u64::MAX) the predicate is exactly seq == prev + 1"

user-provided description — not machine-checked

Formal claim:

```
kani::assume(seq < u64::MAX)
kani::assert(!wrapped)
kani::assume(prev < u64::MAX)
kani::assert(got == (seq == prev.wrapping_add(1)))
```

Harness: `successor_never_wraps_at_the_u64_ceiling`

encoding sha256: e87bb002ddad5cf216c2e7df97470f826a80205820eff0e66ffd16bcf5c19d24

decided by: kani 0.68.0 (bounded)

chain linking is exact successor arithmetic and never wraps at the u64 ceiling

PROVEN 4 / 4

"record N links to record N-1 iff prev_hash equals record N-1's hash AND seq is the exact u64 successor of record N-1's seq; the successor is never taken modulo the u64 ceiling — no seq may link past u64::MAX (the chain cannot wrap) — and on the guarded domain (prev below u64::MAX) the predicate is exactly seq == prev + 1"

user-provided description — not machine-checked

Formal claim:

```
#[ensures(result == (seq == prev + 1u64))]
```

```
#[requires(prev < 18446744073709551615u64)]
```

Target: `links`

`encoding sha256: 92ba29b838925b5024384e389e748475d9bb4e41d1df99beb791ab953efcad62`

provers: alt-ergo closed Coma::vc_checked_add (14 ms); alt-ergo closed Coma::vc_elim_Some (13 ms); alt-ergo closed Coma::vc_links (17 ms); alt-ergo closed Coma::vc_links (16 ms)

decided by: creusot v0.14.0 (65349025c67e) (unbounded) on alt-ergo

Security

Policy	falsegreen-platform
Policy digest	`372d3c74758acb113c1bef14b20b08b326a7703dd57d943247de6541743c0fad`
Result	SATISFIED

Evidence class	Required	Status	Reasons
Dependency/SCA	yes	**PASS**	
Static analysis	yes	**PASS**	
Secrets	yes	**PASS**	
Supply chain	yes	**PASS**	89 components inventoried (evidence, in the signed JSON report)
License policy	yes	**NOT ASSESSED**	no licence policy frozen — the licence facts are inventory, nothing was asserted · 88 licence facts inventoried (evidence, in the signed JSON report)

How to read this table. **PASS** — the evidence was collected and nothing in it violated the frozen policy. **FAIL** — the evidence showed a violation the policy forbids; that alone fails the Job. **INSUFFICIENT EVIDENCE** — the evidence could not be collected, or could not be decided for a reason outside the code (a stale snapshot, an unavailable tool, output the parser could not read). It is never a pass, and it holds the Job at *insufficient_evidence* rather than accepting it. **NOT ASSESSED** — no policy is frozen for this class, so nothing was asserted about it; the facts are recorded as inventory and the class does not hold the Job. The findings table lists only what the frozen policy JUDGED — a graded severity, supply-chain drift, or a licence breaking the frozen policy. Everything else the tools recorded (components, licence facts) is inventory: not a finding, counted on the class row above, kept in the signed JSON report. A finding is **OPEN** unless it is resolved, not affected, or covered by a recorded exception.

Attempt security history (the full per-attempt sections are in the signed JSON report):

Attempt	Result	Open findings	Demonstrated violations
#1	**SATISFIED**	0	—

Witnesses (tool provenance)

```
osv-scanner 2.7.0 (Dependency/SCA) — invocation sha256:
c665185c3c796063a5c13774a3645da26e78f3d0d2fb489d311670b6b068d796 —
a4a5bf077e04f6aeb04a2ea1c0b0852414491d85daa6909080ad17cbf5f7f2e8
```

semgrep 1.181.0 (Static analysis) — invocation sha256: 1483a3a9efd08c1ade64955db6d42f398daa4435b3b58d45ee63bfa0b145a9d8 — a4a5bf077e04f6aeb04a2ea1c0b0852414491d85daa6909080ad17cbf5f7f2e8

gitleaks 8.30.1 (Secrets) — invocation sha256: d15d78be22ceb658f36efce1b182e3fcb0545657ba69593e1b5543796d99882e — a4a5bf077e04f6aeb04a2ea1c0b0852414491d85daa6909080ad17cbf5f7f2e8

syft 1.54.1 (Supply chain) — invocation sha256: 3688baa3543ecf06b574b7f78a6652a6b45e6ead351a9caad0e63781722fdbe3 — a4a5bf077e04f6aeb04a2ea1c0b0852414491d85daa6909080ad17cbf5f7f2e8

dependency-manifest 1 (License policy) — invocation sha256: 8d4f425601989b9bbcbaf67c500a67a793a3b1f3579fd1efaca0d501eeeae4b0 — a4a5bf077e04f6aeb04a2ea1c0b0852414491d85daa6909080ad17cbf5f7f2e8

Tools & provers used

Tool	Version	What it did in this run
osv-scanner	`2.7.0`	Dependency/SCA evidence witness
semgrep	`1.181.0`	Static analysis evidence witness
gitleaks	`8.30.1`	Secrets evidence witness
syft	`1.54.1`	Supply chain evidence witness
dependency-manifest	`1`	License policy evidence witness
kani	`0.68.0`	invariant proofs (1 row)
creusot	`v0.14.0 (65349025c67e)`	invariant proofs (1 row)
alt-ergo, z3	`not recorded (image-pinned)`	authorized provers (frozen contract)

Exports

Full content is embedded in the signed JSON report (report schema v13).

`spdx-3.0 sha256:73788a5f224a5917cbd631dbe04e7af59669fadf9bac5a98e8315c56f61428c3 embedded in the JSON report (66947 bytes)`

`cyclonedx-1.7 sha256:3085b989c3cc1180b77ba2af7489649971ca57bb3f0e69579799160ac30a24a2 embedded in the JSON report (32209 bytes)`

`cyclonedx-1.7-vex sha256:0e80d6ab1d4abbe263baa301053556c687548d3cdfeb60dbd4c35c3d62d0e166 embedded in the JSON report (632 bytes)`

Signature

This report is cryptographically signed: Ed25519 over the canonical report content. The public key is embedded in the report — verification is offline and registry-free (`falsegreen verify-report`).

```
key id: fg-report-9fc1a2c5-134c-4ca1-a0f2-56b2d1bcde67-20261018-1
ed25519 public key: ie4eQdY1lt7O5aCCQzO+/192Yzc4pLQsGyVNqSKtlmA=
ed25519 signature: nPEu6PFm+/BbTMrjHAUN6j/haQLLsslavHQ526mJ6y6nPKsQf+5+gNBKvxn08ISOHL3kEDu1nyH/tp9IvpZtCw==
digest: ac474a614151e7469753ede8d5f1a1794c190e81bdfec280ad425a47435f4ab6
```

This is a presentation of the signed canonical evidence (the JSON artifact). It is not an audit certificate. - FalseGreen v0.15.0